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(a) 

(b) 

(c)  $\mathcal{I}$  metrics

Solution ca) ① density - based clustering

eg. DBSCAN

② Rationale

(1) DBSCAN groups points that are closely packed together (high density) and can isolate region of low density as cluster boundaries.

(2) It doesn't require specifying the number of clusters in advance and can handle complex shapes, like nested rings in Figure 2.

③ procedure of DBSCAN

(1) choose two parameters:

Min Pts: The minimum number of points clustered together for a region to be considered dense.

eps ( $\epsilon$ ): A distance measure that will be used to locate the

points in the neighborhood of any point.

and randomly pick a data point from the dataset as the starting point.

(2) Find recursively all its density connected point and assign them to the same cluster as the core point

(3) Randomly choose another point among the points that haven't been visited in the previous steps and go to step (2)

(4) This process is finished when all points are visited. Those points that don't belong to any cluster are considered as noise.

(b) Centroid based method, such as k-means.

② Rationale

- (1) k-means performs best for roughly spherical, well-separated clusters, where each cluster can be reasonably approximated by a central "mean" point, like Figure 3
- (2) It's computationally efficient and straightforward to implement.

③ Procedure.

- (1) Set the number  $k$  to decide the number of clusters
- (2) Randomly generate  $k$  points or centroids as the initial cluster centres. They can also be randomly selected from the data.
- (3) Assign each data point to their closest centroid, which will form the pre-defined  $k$  clusters

(4) Calculate a new centroid of each cluster

(5) Repeat steps 3-4 until there is no change in the centroids and the assignment of sample.

(c) ① Silhouette coefficient measures how well each sample fits into its assigned cluster.

② Dunn index

The Dunn index aims at quantifying the compactness and variance of the clustering

③ Davies - Bouldin index

Davies - Bouldin index evaluates the goodness of clustering based on within-cluster dispersion and between-cluster separation

④ Calinski - Harabasz index

The Calinski - Harabasz index is a measure of how similar a sample is to

its own cluster (cohesion) compared to other cluster (separation).

### ⑤ Rand index

Rand index compute a similarity measure between two clustering methods by considering all pairs of samples and counting pairs that are assigned in the same or different clusters across two different methods.